

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a9db2eb8603e0b45a.jsonl`  
- SHA-256: `cf083e13f87c3b5a32a080b34d2dc0494e9f0e89caa15e774337c4a7ae7ac972`  
- Source modified: `2026-05-31T04:26:20+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:25:19.321Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"On table extraction accuracy (table edit-distance, lower is better), the PP-StructureV3 OCR+table-structure pipeline outperforms large general-purpose VLMs and pipeline competitors: it scores 0.159 (EN) / 0.109 (ZH) on Table Edit, versus Qwen2.5-VL-72B at 0.341/0.262, GOT-OCR at 0.459/0.52, Marker-1.2.3 at 0.619/0.685, MinerU-0.9.3 at 0.18/0.344, and Mathpix at 0.243/0.32. This supports a deterministic-pipeline-over-VLM recommendation for dense tables."

****Source:**** <https://github.com/PaddlePaddle/PaddleOCR/blob/main/docs/version3.x/algorithm/PP-StructureV3/PP-StructureV3.en.md> (primary)

****Supporting quote:**** "PP-StructureV3 scores: ... Table Edit: EN 0.159, ZH 0.109 ...

Qwen2.5-VL-72B: ... Table 0.341 (EN), 0.262 (ZH) ... GOT-OCR: ... Table 0.459 (EN), 0.52 (ZH) ... Marker-1.2.3: ... Table 0.619 (EN), 0.685 (ZH)"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:25:21.981Z)

I'll verify this claim about PP-StructureV3's table extraction benchmarks. Let me start by examining the primary source and searching for corroborating/contradicting evidence.

3. assistant (2026-05-31T04:25:25.205Z)

I'll fetch the primary source first and search for independent corroboration in parallel.

4. user (2026-05-31T04:25:31.305Z)

PP-StructureV3 Table Edit Distance Benchmark Results

Based on the OmniDocBench dataset, here are the exact "Table Edit" distance scores:

Method	EN	ZH
PP-StructureV3	0.159	**0.109**
Qwen2.5-VL-72B	0.341	0.262
GOT-OCR	0.459	0.52
Marker-1.2.3	0.619	0.685
MinerU-0.9.3	0.18	0.344
Mathpix	0.243	0.32

****Source:**** The document references "OmniDocBench: Benchmarking Diverse PDF Document Parsing with Comprehensive Annotations" (arxiv.org/abs/2412.07626). No specific document date is provided in the content, though the document is hosted in PaddleOCR's main branch repository.

PP-StructureV3 demonstrates superior table recognition performance, particularly in Chinese text with the lowest error rate at 0.109.

5. user (2026-05-31T04:25:33.412Z)

Web search results for query: "PP-StructureV3 OmniDocBench Table Edit distance 0.159 Qwen2.5-VL-72B benchmark"

Links: [{"title": "GitHub - marklabz/omnidocbench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation · GitHub", "url": "https://github.com/marklabz/omnidocbench"}, {"title": "GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation", "url": "https://github.com/opendatalab/OmniDocBench?tab=readme-ov-file"}, {"title": "GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation · GitHub", "url": "https://github.com/opendatalab/OmniDocBench"}, {"title": "OmniDocBench v1.0 Benchmark", "url": "https://www.emergentmind.com/topics/omnidocbench-v1-0"}, {"title": "DocPTBench: Parsing & Translation Benchmark", "url": "https://www.emergentmind.com/topics/docptbench"}, {"title": "Real5-OmniDocBench: A Full-Scale Physical Reconstruction Benchmark for Robust Document Parsing in the Wild", "url": "https://arxiv.org/html/2603.04205v1"}, {"title": "OmniDocBench v1.5 Benchmark", "url": "https://www.emergentmind.com/topics/omnidocbench-v1-5"}, {"title": "Fine-Tuning Vision-Language Models for Markdown Conversion of Financial Tables in Malaysian Audited Financial Reports", "url": "https://arxiv.org/pdf/2508.05669"}, {"title": "OmniDocBench Leaderboard: Document Parsing SOTA | CodeSOTA | CodeSOTA", "url": "https://www.codesota.com/ocr/benchmark/omnidocbench"}]

Based on the search results, I found information about PP-StructureV3 and its performance on the OmniDocBench benchmark.

PP-StructureV3 achieved a Table Edit Distance of 0.073 on the OmniDocBench benchmark, not 0.159 as mentioned in your query.

The search results show that PP-StructureV3 was added to OmniDocBench model evaluation on July 31, 2025. However, the specific edit distance value of 0.159 you mentioned does not appear in the benchmark results I found. The 0.073 value for PP-StructureV3 represents a very strong performance on table parsing tasks.

For context, the Qwen2.5-VL-72B model achieved a Table Edit Distance of 0.094 on the same benchmark, which is slightly higher (worse) than PP-StructureV3's performance.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:25:37.262Z)

The primary source confirms the exact numbers in the claim verbatim. But the web search surfaced a critical discrepancy: a separate OmniDocBench leaderboard reports PP-StructureV3 at 0.073 Table Edit and Qwen2.5-VL-72B at 0.094 ? radically different numbers and, crucially, a much narrower gap. This suggests the PaddleOCR-published numbers may be from an older/self-reported benchmark version. Let me investigate the discrepancy and check the OmniDocBench versioning.

7. user (2026-05-31T04:25:43.545Z)

OmniDocBench Table Edit Distance Scores

Based on the leaderboard content provided, here's what I found:

****Available scores:****

- ****PP-StructureV3****: 0.145 (EN only, from OmniDocBench v1.0)
- ****Qwen2.5-VL 72B****: 0.226 (EN only, from OmniDocBench v1.0)

****Not found on this leaderboard:****

GOT-OCR, Marker, MinerU, and Mathpix do not appear in the Table Edit Distance rankings shown.

****Benchmark version & update date:****

This leaderboard contains mixed versions. The composite score section references "OmniDocBench v1.5," while the text edit distance section cites both v1.0 and v1.5 results. The page was "Updated 2026-05-21" according to the header.

****Note:**** The leaderboard distinguishes between different metrics?the Table TEDS section shows only 4 results total, with paddleocr-vl leading at 93.52. The text edit distance scores you may be seeking appear in separate sections from the table structure metrics.

8. user (2026-05-31T04:25:46.412Z)

OmniDocBench Table Recognition Scores

Based on the provided documentation, here's what's available:

Table TEDS Scores (v1.0)

The benchmark includes a component-level table recognition evaluation table showing overall TEDS scores, but ****specific breakdowns by EN/ZH language are not provided in the excerpt for most models****:

- ****PaddleOCR****: 73.6 overall
- ****GOT-OCR****: 74.9 overall
- ****Mathpix****: Not listed in table recognition evaluation
- ****MinerU, Qwen2.5-VL-72B, PP-StructureV3, Marker****: Not included in the table recognition results section

Version and Timeline

The benchmark distinguishes between versions:

- ****v1.0****: Original release (table recognition evaluation available)
- ****v1.5****: Updated version with enhanced algorithms and expanded dataset
- ****v1.6****: Latest version with "Multi-Granularity Adaptive Matching"

****Update timing****: The most recent updates mentioned are from 2026 (future dates in the document, likely documentation errors), with substantive changes noted in March-April 2025.

Key Limitation

The document focuses primarily on end-to-end evaluation scores rather than isolated table recognition metrics for the models you mentioned. PP-StructureV3 and others appear in end-to-end

leaderboards but lack dedicated table-only evaluation data in this excerpt.

9. user (2026-05-31T04:25:48.402Z)

Web search results for query: "PP-StructureV3 0.073 Table Edit OmniDocBench Qwen2.5-VL-72B 0.094 leaderboard"

Links: [{"title": "GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation · GitHub", "url": "https://github.com/opendatalab/OmniDocBench"}, {"title": "open-llm-leaderboard/Qwen_Qwen2-VL-72B-Instruct-details · Datasets at Hugging Face", "url": "https://huggingface.co/datasets/open-llm-leaderboard/Qwen_Qwen2-VL-72B-Instruct-details"}, {"title": "DocVQA Benchmark Leaderboard", "url": "https://llm-stats.com/benchmarks/docvqa"}, {"title": "Real5-OmniDocBench: A Full-Scale Physical Reconstruction Benchmark for Robust Document Parsing in the Wild", "url": "https://arxiv.org/html/2603.04205v1"}, {"title": "OmniDocBench Leaderboard: PDF Document Parsing SOTA | CodeSOTA | CodeSOTA", "url": "https://www.codesota.com/browse/computer-vision/document-parsing/omnidocbench"}, {"title": "OmniDocBench Leaderboard ? v1.5 | IDP Leaderboard", "url": "https://www.idp-leaderboard.org/benchmarks/omnidocbench"}, {"title": "Qwen2.5-VL Technical Report", "url": "https://arxiv.org/pdf/2502.13923"}]

Based on the search results, I found information about the models and benchmark you're asking about:

PP-StructureV3 and Qwen2.5-VL-72B are both models that have been evaluated on OmniDocBench, a comprehensive document parsing benchmark. OmniDocBench includes 1651 PDF pages covering 10 document types, 5 layout types, and 5 language types, with rich annotations including 28 block-level annotations such as text paragraphs, headings, and tables.

PP-StructureV3 and Qwen2.5-VL-72B are among the officially aligned baseline models verified in the original OmniDocBench report. The specific performance metrics you mentioned (PP-StructureV3 at 0.073 for Table Edit and Qwen2.5-VL-72B at 0.094) appear to be part of the OmniDocBench leaderboard results, though the detailed performance breakdown for these specific metrics is not fully visible in the search results.

If you need more detailed information about the specific scores and metrics, you may want to visit the official OmniDocBench repository or leaderboard directly at GitHub or the official benchmark sites.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

10. assistant (2026-05-31T04:25:53.190Z)

I'm seeing multiple different numbers for the same models on the same benchmark, which indicates version drift. Let me reason about what I have and check the IDP leaderboard plus the OmniDocBench README results table directly for the canonical figures.

The picture so far:

- PaddleOCR's self-published doc: PP-StructureV3 EN 0.159, Qwen2.5-VL-72B 0.341
- codesota: PP-StructureV3 EN 0.145, Qwen2.5-VL-72B 0.226 (attributed to "v1.0")
- A separate result set: PP-StructureV3 0.073, Qwen2.5-VL-72B 0.094

These are wildly inconsistent. Let me get the authoritative OmniDocBench numbers and check the IDP leaderboard.

11. user (2026-05-31T04:25:59.720Z)

OmniDocBench Table TEDS Scores

Based on the leaderboard content provided, I cannot find Table TEDS (TEDS?) scores for the models you requested: PP-StructureV3, Qwen2.5-VL-72B, GOT-OCR, Marker, MinerU, or Mathpix. These

models do not appear in the rankings table shown.

****Benchmark version:**** This is "OmniDocBench v1.5," which consists of "1,355 pages from papers, books, slides, exams, newspapers, and magazines."

****Best performer on tables:**** Nanonets OCR2+ ranks highest on the Table TEDS metric with a score of 79.1, placing third overall.

The leaderboard displays 29 evaluated models, but the six you mentioned are not among those listed in this version.

12. user (2026-05-31T04:26:02.278Z)

End-to-End Evaluation Results

Based on the document provided, I cannot find a table showing separate "EN" and "ZH" edit distance values for the specific methods you requested (PP-StructureV3, Qwen2.5-VL-72B, GOT-OCR, Marker, MinerU).

The main end-to-end evaluation table in the document displays ****Text Edit Distance**** as a single "?" column (lower is better), not separated by language. For example:

- ****Marker****: Text Edit Distance = 0.157
- ****MinerU-2.5****: Text Edit Distance = 0.045
- ****MinerU2.5-Pro****: Text Edit Distance = 0.036

However, the document notes that in prior versions (v1.0 and v1.5), "The leaderboard removed the Chinese/English grouping, now calculating the average score across all pages."

The document does not provide the granular language-separated metrics you're seeking in its current evaluation tables.

13. user (2026-05-31T04:26:05.688Z)

Web search results for query: "OmniDocBench v1.0 vs v1.5 different scores PP-StructureV3 MinerU benchmark version"

Links: [{"title":"GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation · GitHub","url":"https://github.com/opendatalab/OmniDocBench"}, {"title":"Many pipelines that were evaluated in v1 not evaluated in v1.5 · Issue #120 · opendatalab/OmniDocBench","url":"https://github.com/opendatalab/OmniDocBench/issues/120"}, {"title":"Real5-OmniDocBench: A Full-Scale Physical Reconstruction Benchmark for Robust Document Parsing in the Wild","url":"https://arxiv.org/html/2603.04205v1"}, {"title":"OmniDocBench v1.0 Benchmark","url":"https://www.emergentmind.com/topics/omnidocbench-v1-0"}, {"title":"OmniDocBench v1.5 Benchmark","url":"https://www.emergentmind.com/topics/omnidocbench-v1-5"}, {"title":"MinerU2.5-Pro: Pushing the Limits of Data-Centric Document Parsing at Scale","url":"https://arxiv.org/pdf/2604.04771"}, {"title":"OmniDocBench Leaderboard: PDF Document Parsing SOTA | CodeSOTA | CodeSOTA","url":"https://www.codesota.com/browse/computer-vision/document-parsing/omnidocbench"}]

Based on the search results, I can provide information about OmniDocBench version differences and PP-StructureV3/MinerU performance:

Version Differences

The main branch of evaluation code and dataset have been updated to Version v1.5, with the v1.0 available in a separate branch (v1_0).

OmniDocBench v1.5 is a comprehensive benchmark for diverse PDF document parsing, designed to address limitations in dataset diversity, annotation granularity, and evaluation methodologies present in previous benchmarks.

PP-StructureV3 and MinerU Scores

According to the available benchmark data, here are the scores for these models:

In the newer version, MinerU achieves 0.133 on one metric and 90.67 and 61.5 on others, while PP-StructureV3 achieves 0.145 and 86.73.

For context on performance differences: Pipeline tools like MinerU outperform vision-LLMs for English text and formulas, with pipelines excelling for academic papers and financial reports while VLMs outperform on slides and handwritten notes.

Additionally, a Hard subset was introduced to establish the more discriminative OmniDocBench v1.6 protocol, indicating the benchmark has continued to evolve beyond v1.5.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

14. assistant (2026-05-31T04:26:15.124Z)

I now have enough to understand the discrepancy. Let me reason carefully about what the claim actually says versus what is true.

The claim is specifically about **the PaddleOCR-published benchmark numbers** and whether they support a "deterministic-pipeline-over-VLM" recommendation. Let me separate two distinct questions:

Question A: Is the quote accurately reproduced? Yes. WebFetch on the primary source confirmed all six methods' numbers verbatim (PP-StructureV3 0.159/0.109, Qwen2.5-VL-72B 0.341/0.262, GOT-OCR 0.459/0.52, Marker-1.2.3 0.619/0.685, MinerU-0.9.3 0.18/0.344, Mathpix 0.243/0.32). The numbers and ordering are real and the claim reproduces them correctly.

Question B: Do these numbers reliably support the broader conclusion? This is where I'm finding serious problems. The same models on the same nominal benchmark (OmniDocBench) show drastically different Table Edit scores across versions:

- PaddleOCR doc (this claim): PP-StructureV3 **0.159**, Qwen2.5-VL-72B **0.341** (gap ~2.1x)
- codesota/v1.0: PP-StructureV3 **0.145**, Qwen2.5-VL-72B **0.226** (gap ~1.6x)
- Another result set: PP-StructureV3 **0.073**, Qwen2.5-VL-72B **0.094** (gap ~1.3x)

Let me verify the version/self-reporting issue and the "MinerU-0.9.3" staleness, since MinerU is now at 2.5/2.5-Pro. Let me also check the OmniDocBench issue about pipelines not re-evaluated.

15. user (2026-05-31T04:26:18.701Z)

You've hit your session limit · resets 1:30pm (Asia/Calcutta)

16. user (2026-05-31T04:26:20.111Z)

You've hit your session limit · resets 1:30pm (Asia/Calcutta)

17. user (2026-05-31T04:26:20.306Z)

Web search results for query: "MinerU 2.5 OmniDocBench table edit distance vs PP-StructureV3 2025 latest results VLM outperform pipeline"

You've hit your session limit · resets 1:30pm (Asia/Calcutta)

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

18. assistant (2026-05-31T04:26:20.691Z)

You've hit your session limit · resets 1:30pm (Asia/Calcutta)